Module 2: Overfishing

```r
library(tidyverse)
```

Use of GitHub

Link to your forked GH repository: https://github.com/nikhilshah7/CLES131-module1-climate

Use of Quarto

Link to your .qmd file: https://github.com/nikhilshah7/CLES131-module1-climate/blob/main/module1.qmd

Overexploitation of global fisheries

We will examine the state of global fisheries and seek to reproduce one of the most widely cited examples of species collapse in order to examine the evidence behind an influential paper on global fisheries, Worm et al. 2006. However, rather than using the 1950-2003 dataset available to Worm and colleagues in 2006, we will be drawing from more recent stock assessment data to see how the trends have fared, specifically in Atlantic cod.

We also explore how to understand and manipulate tabular data using relational database concepts. Instead of working with a single independent dataframe, as in Module 1, we will be working with a large relational database consisting of tables of different dimensions, but are related through one or more IDs.

The Database

We will use data from the RAM Legacy Stock Assessment Database. Note that the database files are available through Zenodo; please download and unzip the file, moving the .RData file only into your own `data/` folder within this project.

Use the code below to load up the tables in this database.

```
load("data/DBdata[asmt][v4.66].RData")
```

The above function loads all 60 tables of this database, which is a lot! So step one of the data science process is exploring the structure of the database and figuring out how the tables are related.

A selected set of documentation have been provided in `documentation/`. Open up the Database Quickstart Guide for reference as you work through the problems below.

Q1 (1 point)

One of the main metadata tables is 'stock':

- How many stocks are there and how many species do they represent?

```
nrow(stock)
```

```
[1] 1514
```

```
# 1514

length(unique(stock$tsn))
```

```
[1] 398
```

```
# 398
```

- What are the top three most common stocks? (Okay to use common names; these are standardized for fish)

```
stock |>
  group_by(commonname) |>
  summarize(n_stocks = n()) |>
  arrange(desc(n_stocks))
```

```
# A tibble: 410 x 2
   commonname      n_stocks
   <chr>              <int>
 1 Pink salmon          170
 2 Sockeye salmon       104
 3 Chum salmon           99
 4 Norway lobster        38
 5 Atlantic cod          28
 6 Herring               24
 7 Anchovy               19
 8 Snow crab             19
 9 Northern shrimp       17
10 Haddock               14
# i 400 more rows
```

```
# Pink salmon (170)
# Sockeye salmon (104)
# Chum salmon (99)
```

- In your own words, describe the unit of observation in the 'stock' table and some associated variables that describe it.
- Each row describes one fish stock, a population of a species of fish in one given location. Variables relate to the species of fish (tsn, scientificname, commonname), stock location (areaid, region, primary_country), and stock status (state).

Q2 (1 point)

The next main metadata table is 'assessment':

- What is the unit of observation in the 'assessment' table? How is this related to the unit of observation in the 'stock' table?

```
# Each row is one assessment of one stock.

length(unique(assessment$stockid))
```

```
[1] 1514
```

```
# Each stock is assessed at least once in this table (1514 stocks total). Stocks may be asses
```

- It seems like ‘stockid’ and ‘assessyear’ might uniquely identify each assessment. Is this the case? If not, why not?

```
assessment |>
  group_by(stockid, assessyear) |>
  summarize(n = n()) |>
  arrange(desc(n))
```

```
`summarise()` has grouped output by 'stockid'. You can override using the
`.groups` argument.
```

```
# A tibble: 3,449 x 3
# Groups:   stockid [1,514]
   stockid           assessyear     n
   <chr>             <chr>      <int>
 1 ANCHIXa           1989-2016      2
 2 BLINGVa-XIV       2000-2016      2
 3 BRILL2232         1970-2016      2
 4 COD1IN            1911-2016      2
 5 CODBA2224         1993-2016      2
 6 CODBA2532         1965-2016      2
 7 CODIIIaW-IV-VIId 1962-2016      2
 8 DAB2232           1970-2016      2
 9 EFLOUN2223        1973-2016      2
10 EFLOUN2425        1973-2016      2
# i 3,439 more rows
```

```
# No, some stocks were assessed multiple times for the same time period, such as anchovy in S
```

- Which variable(s) might you use to join ‘stock’ and ‘assessment’?
- stockid, as it seems to be the only shared variable between the two datasets.

Q3 (1 point)

In section B of the Database Quick Guide, additional metadata tables are listed in section B. With your data exploration skills, complete the following table:

B table	Associated table	join column from B	join column from other
area	stock	areaid	areaid

B table	Associated table	join column from B	join column from other
assessmethod	assessment	methodshort	assessmethod
assessor	assessment	assessorid	assessorid
management	stock	country	primary_country
taxonomy	stock	tsn	tsn

*Note that assessmethod is not a data.frame, which was perhaps an oversight; consider turning it into one for the ease of using tidyverse functions.

```
assessmethod <- data.frame(assessmethod)
```

Q4 (1 point)

There are two main data tables described in section C, but their key attributes are difficult to understand. Join the two main data tables with their respective metrics tables (section D):

- Briefly describe the contents of each joined table.
- What is accomplished by joining these tables?

Overexploitation of Atlantic cod

This database is clearly complex and contains a lot of information. We've grappled with its high dimensionality by exploring and joining tables above. Next, we will take a "bottom-up" approach and evaluate a single species, Atlantic cod. Read this book chapter on Atlantic cod, which serves as a reference for the following questions.

Q5 (1 point)

The collapse of the Atlantic cod fisheries is well documented. Let's see if we can visualize that decline with this database.

- How many unique fish stocks comprise "Atlantic cod" and what regions do they come from?

Create a vector of the 'stockid' associated with "Atlantic cod". Then turn your attention to the previously-joined timeseries/tsmetrics table and create a subsetted dataframe with only observations of "Atlantic cod" that have a non-NA value. This table reports many years and many metrics, while we are primarily interested in the years since 1950.

- Which metrics are most commonly reported for Atlantic cod in this timeframe and what do they mean?

Q6 (1 point)

Make a timeseries figure for each stock of Atlantic cod. Rather than coloring by the identity of the stock, color by the region it comes from (this may require joining). Use `facet_*()` to plot both metrics in a single code chunk. Connect the points for each stock and focus on the period since 1950. Update until the figure is easily interpretable.

- What happened to Atlantic cod stocks during the period 1950-present? What information does each metric convey?

Q7 (1 point)

We will recreate a slightly more complex version of a figure from the Millennium Ecosystem Assessment (Fig. 11).

Selecting a metric that can be added across stocks, calculate the total fish landings for each region of Atlantic cod. Make a plot similar to Fig. 11 and color by region.

Which region does Fig. 11 from the MEA correspond to?

Q8 (1 point)

Under the Management Policies section of the book chapter, the authors summarize a few major efforts to restore and protect stocks of Atlantic cod. Focusing only on the US and Canada, update your plot from above to include vertical lines to represent the timing of least three major events and label them clearly on your plot. Doing so will likely involve creating a separate dataframe and using it in one of your ggplot layers (one possible argument within `aes()` is `linetype`). Facet the plot by region and update until it is easily interpretable.

Q9 (2 points)

What happened to Atlantic cod stocks for each region following each piece of legislation or management plan, and why do you think this was so? Your multi-paragraph answer should be informed by content from the book chapter; additional sources are optional, and all sources should be cited.